Reg. No.															

B.TECH DEGREE EXAMINATION, NOVEMBER 2024

First and Second Semester

21CYB101J - CHEMISTRY

(For the candidates admitted from the academic year 2021-2022 to 2023-2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
1.	Which factor is primarily responsible for the difference in magnetic properties between high-spin and low-spin octahedral complexes?	1	2	1	4
	A) Ligand field strength B) Coordination number C) Oxidation state of the metal D) Nature of the counter-ion				
2.	According to the Hard and Soft Acids and Bases (HSAB) principle, which of the following pairs will have the strongest ionic interaction?	1	2	1	4
	A) Soft acid and hard base B) Hard acid and soft base C) Soft acid and soft base D) Hard acid and hard base				
3.	The second ionisation energy is always higher than the first ionization energy because the_____.	1	2	1	4
	A) electron is attracted more by the core electrons B) electron is more tightly bound to the nucleus in an ion C) becomes more stable attaining the octet or duplet configuration D) atomic radii is large				
4.	The CFSE for a high spin d4 octahedral complex is_____.	1	2	1	4
	A) $-0.6\Delta_o$ B) $-1.8\Delta_o$ C) $-1.6\ \Delta_o+P$ D) $-1.2\Delta_o$				
5.	Which of the following is the correct criterion for a spontaneous process?	1	2	2	3
	A) ΔS system-ΔS surroundings B) ΔS surroundings>0 only C) ΔS system+ΔS surroundings >0 D) ΔS system>0 only				
6.	The correct statement about cell potential is _____.	1	2	2	3
	A) sum of the electrode potentials of the cathode and anode B) difference between the electrode potentials of the cathode and anode C) half of the sum of the electrode potentials of the cathode and anode D) twice the difference between the electrode potentials of the cathode and anode				
7.	In Pourbaix diagram, the redox reaction, $Fe^{2+}+2e^{-}\rightarrow Fe(s)$ is _____.	1	2	2	3
	A) solvent dependent B) solvent independent C) pH dependent D) pH independent				
8.	Which corrosion product is volatile in nature?	1	2	2	3
	A) MoO_3 B) Fe_2O_3 C) Fe_3O_4 D) $CuCO_3$				

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9.	The potential energy of n-butane is minimum for_____.	1	3	3	4
	A) Skew conformations B) Staggered conformations C) Eclipsed conformations D) Gauche				
10.	Which of the following stereoisomers will rotate plane-polarized light in opposite directions?	1	3	3	4
	A) Diastereomers B) Enantiomers C) Geometrical isomers D) Conformational isomers				
11.	A molecule with a plane of symmetry is_____.	1	2	3	4
	A) Always optically active B) Always a diastereomer C) Achiral D) Chiral				
12.	Which of the following is NOT an optically active compound?	1	2	3	4
	A) 1,7- Dicarboxylic Spiro Cycloheptane B) 1,3- Diphenylpropadiene C) Glyceraldehyde D) meso-tartaric acid				
13.	The type of linkage present in poly urethane is_____.	1	2	4	4
	A) Glycosidic linkage B) Phospho diester linkage C) Ester linkage D) Amide linkage				
14.	Which of the following is NOT a characteristic of thermosetting polymers?	1	2	4	4
	A) They cannot be remelted after curing B) They are typically brittle and hard C) They can be reshaped by heating D) They form cross-linked structures during curing				
15.	Which of the following is a co-polymer?	1	1	4	4
	A) Polythene B) Polystyrene C) PVC D) Buna-S-Rubber				
16.	Which among the following polymers have lowest solubility?	1	2	4	4
	A) epoxy resin B) polyethylene C) polystyrene D) PET				
17.	In storage battery plates the matrix commonly used is_____.	1	2	5	5
	A) Aluminium B) Lead C) Silver D) Copper				
18.	Which of the following material is commonly used in ceramic matrix composites?	1	2	5	5
	A) Carbon B) Aluminum C) Silicon carbide D) Rubber				
19.	Kevlar is a_____type of material.	1	2	5	5
	A) Glass B) Thermoplastic C) Whisker D) Polymer				
20.	The Bragg's equation for diffraction of X-rays is_____.	1	2	5	5
	A) $n\lambda = 2d^2\sin\theta$ B) $n\lambda = 2d\sin\theta$ C) $n\lambda = 2d\sin^2\theta$ D) $n\lambda = d^2\sin\theta$				

PART - B (5 x 8 = 40 Marks) **Marks BL CO PO**

Answer **ALL** the Questions

21 a.	Describe the salient features of Crystal Field Theory (CFT) and write its limitations.	8	2	1	4
	(OR)				
b.	How does the geometry of a transition metal complex, the nature of the ligands, the metal's oxidation state, and the spectrochemical series influence its optical properties and color?	8	2	1	4

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22 a.	Derive Gibbs-Helmholtz equation of free energy and write any one application.	8	2	2	3
	(OR)				
b.	Draw a neat sketch of Pourbaix diagram for iron and explain.	8	2	2	3
23 a.	Explain Cahn-Ingold Prelog rules to determine R/S configuration on a chiral center.	8	2	3	4
	(OR)				
b.	Explain with a neat diagram about the conformational analysis of n-butane.	8	2	3	4
24 a.	Discuss the synthesis, properties and applications of PTFE and Polystyrene.	8	2	4	4
	(OR)				
b.	Explain n-doping and p-doping in conducting polymers.	8	2	4	4
25 a.	Provide a graphical illustration of the stress-strain relationship in solids and explain it in detail.	8	2	5	5
	(OR)				
b.	Explain the preparation, properties and uses of Fiber Reinforced Polymer and Metal Matrix composite.	8	2	5	5

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Questions

		Marks	BL	CO	PO
26	i)Describe with suitable examples, the structural isomerism in coordination compounds. (8 Marks) ii) Derive Nernst equation for the redox potential of a reversible reaction and write its advantages. (7 Marks)	15	2	1	4
27	i) Explain the principle and instrumentation of X-ray Photo Electron Spectroscopy. (10 Marks) ii) Discuss in detail about SN^1 reaction mechanism. (5 marks)	15	2	5	5

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THE HELPERS